
PicoLab

A visual STEM learning coach that turns mistakes into learning signals.

Hackathon Submission · AI x STEM Education · GitHub: <https://github.com/Alehcs/PicoLab>

PURPOSE

PicoLab helps students solve STEM problems through a guided learning loop: problem intake, confirmation, step-by-step solving, visual explanation, personalized practice, and progress tracking.

Its core idea is simple: mistakes should become learning signals, not failures.

PROBLEM

Students often get a numeric answer partially correct while still misunderstanding the concept, formula, graph, or unit behind it. Many tutoring tools provide answers, but do not clearly diagnose the type of mistake or convert it into a practice path.

SOLUTION — THE STUDENT WORKFLOW

1. Add or scan a STEM problem
2. Confirm values, units, target variable, and formula
3. Solve step-by-step in Smart Notebook
4. Detect learning signals
5. Open a contextual Visual Lab explanation
6. Practice through missions
7. Track progress in Growth Map and Roadmap

KEY FEATURES

- **Add Problem** — type, scan, or formula-based input
- **Scan & Confirm** — structured review before solving
- **Smart Notebook** — step-by-step feedback
- **Learning Signals** — unit, formula, graph, algebra, concept, and reading signals
- **Visual Lab** — interactive explanations linked to the detected signal
- **Practice Missions** — daily challenge, recommended practice, and random mission
- **Growth Map** — tracks recurring learning signals
- **Roadmap** — turns signals into a learning path
- **Ask Pico** — contextual coaching drawer with mock/fallback behavior

TECHNICAL IMPLEMENTATION

- React + Vite + TypeScript
- Tailwind-based design system
- Express + TypeScript mock backend
- Backend-first API client with local fallback
- Deterministic diagnostic engine
- localStorage / sessionStorage persistence
- Provider-ready architecture for future real AI integration

DEMO FLOW

Problem → **Confirm** → **Smart Notebook** →
Learning Signal → **Visual Lab** → **Practice Mission** →
Growth Map → **Roadmap**

PROJECT STATUS

PicoLab is a functional hackathon prototype. The current demo uses deterministic mock/backend logic and a canonical physics problem to demonstrate the full learning loop. The architecture is ready for future expansion with real AI providers, broader STEM domains, and richer visual simulations.